

»» Chapter IX ««

THE GAMMA AND ZETA FUNCTIONS. PRIME NUMBER THEOREM

PREFATORY REMARKS

This chapter—the final one of the first part of the book—extends the examination of the gamma function and Riemann zeta function initiated in the exercises of §3, Ch. VIII. In part, our object is to prepare the way for a proof of the Prime Number Theorem of Hadamard and de la Vallée Poussin, which states:

With $\pi(x)$ denoting the number of positive primes not exceeding the positive number x , we have

$$\lim_{x \rightarrow \infty} \pi(x) \frac{\log x}{x} = 1.$$

This important theorem of number theory is of great interest at the present stage of the student's experience in complex function theory because the preparation for its proof brings into play a number of basic concepts and results developed thus far in the book. We cite the following: the Cauchy integral theorem, the residue theorem, the identity theorem for meromorphic functions, analytic logarithms, infinite products, the Hadamard representation theorem.

The gamma function, introduced in the eighteenth century by Euler, has been the subject of numerous investigations. The Riemann zeta function is of basic importance in analytic number theory. The celebrated conjecture of Riemann concerning the distribution of its zeros has for a century challenged the efforts of some of the most eminent mathematicians. We shall see that properties of the gamma function enter the study of the Riemann zeta function in an essential way. Prominence will be given in this chapter to the elegant characterization of $\Gamma \mid \mathbf{P}$ (we recall that $\mathbf{P}$ is the set of positive reals) due to Harald Bohr and the beautiful treatment of the gamma function based on Bohr's characterization given by E. Artin in a short monograph now available in English translation [7].

It would be an overwhelming task to survey the literature on the gamma and Riemann zeta functions and one not consonant with our objectives. For the gamma function cf. [31], [77], [130]; for the Riemann zeta function cf. [118].

Accounts of the Prime Number Theorem are given by Dinghas [38] and Valiron [120] in their treatises. Of course, treatments are to be found in works on analytic number theory.

1. THE GAMMA FUNCTION

We recall that the gamma function Γ is the reciprocal of the entire function g given by

$$g(z) = e^{\gamma z} z \prod_0^{\infty} E_1 \left[\frac{z}{-(n+1)} \right], \quad (1.1)$$

where γ is the Euler constant. At this moment we see that $g(z) = e^{\gamma z} P(z)$, where P is the Weierstrass canonical product associated with the divisor ∂ taking the value 1 at the nonpositive integers and the value 0 elsewhere in $\mathbf{C}$ and that the genus of ∂ , $\kappa(\partial)$, and $\alpha(g)$ are all equal to 1.† We recall also that Γ satisfies the functional equation

$$\Gamma(z+1) = z \Gamma(z). \quad (1.2)$$

Let $f = \log \circ (\Gamma \mid \mathbf{P})$. It follows from (1.1) that $f''(x) > 0$, x positive. In fact,

$$f''(x) = \sum_0^{\infty} (n+x)^{-2}.$$

† Only the assertion $\alpha(g) = 1$ calls for attention. That $\alpha(g) \leq 1$ follows from the facts that $\alpha(P) = 1$ and that the order of the product of entire functions f_1, f_2 does not exceed the maximum of the orders of f_1 and f_2 . In the other direction we know that $\alpha(g) \geq \kappa(\partial)$ by Theorem 6.1, Ch. VIII. Hence $\alpha(g) = 1$.

Further $g(1) = 1$ as we see from (1.1) on considering the partial products, account being taken of the definition of the Euler constant. Trivially, it follows that $\Gamma(1) = 1$. H. Bohr showed that $\Gamma|_{\mathbf{P}}$ is characterized by the conditions: (1) it takes positive values, 1 at 1, (2) it satisfies the functional equation (1.2) for positive z , (3) its logarithm (in the sense of real analysis) has a second derivative taking positive values everywhere on $\mathbf{P}$; and he showed that for sufficiency the condition (3) may be replaced by the requirement that the logarithm of the function in question be *convex*. We shall establish the sufficiency part of the Bohr theorem after we treat the notion of the convexity of a real-valued function of a real variable.

We have already treated convexity on $\mathbf{C}$. For $\mathbf{R}^2$ we merely paraphrase (for that matter the notion is available for vector spaces over $\mathbf{R}$ or $\mathbf{C}$) and say that $E \subset \mathbf{R}^2$ is *convex* provided that

$$(a(1-t) + ct, b(1-t) + dt) \in E,$$

$0 \leq t \leq 1$, whenever $(a, b), (c, d) \in E$. Given $f: C \rightarrow \mathbf{R}$, where C is a connected subset of $\mathbf{R}$, we say that f is a *convex function* provided that the set

$$E(f) = \{(x, y): x \in C, y \geq f(x)\}$$

is convex.† The following facts are easily verified: (1) f is a convex function if and only if

$$f(x) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b),$$

$a \leq x \leq b$, where $a, b \in C$, $a < b$. (2) f is a convex function if and only if

$$\frac{f(b) - f(a)}{b - a} \leq \frac{f(c) - f(b)}{c - b},$$

$a < b < c$, where $a, b, c \in C$. (3) If f'' exists on C , then f is a convex function if and only if $f''(x) \geq 0$, $x \in C$.

EXERCISE

1.1. Verify the above assertions. For (3) the mean-value theorem of the differential calculus and the definition of derivative are pertinent. Show that an equivalent condition is obtained when “ b ” is replaced by “ a ” on the right side of the inequality in (2).

Show that a convex function is continuous at each interior point of its domain.

† It is admittedly a minor point, but a convex function need not be a convex subset of $\mathbf{R}^2$. Since confusion will not arise, there is no need to break with usage.

The following theorem of Bohr yields the sufficiency of the Bohr characterization of the gamma function.

Theorem 1.1: *Given $f: \mathbf{P} \rightarrow \mathbf{R}$ a convex function and satisfying (1) $f(1) = 0$, (2) $\exp \circ f(x+1) = x \exp \circ f(x)$, x positive. Then $\Gamma|_{\mathbf{P}} = \exp \circ f$.*

PROOF: Suppose that $1 < x < 2$. Let $n \in \mathbf{N}$. From (2) we obtain

$$f(x+n+1) - f(x) = \sum_0^n \log(x+k).$$

Hence, since $f(n+1) = \log(n!)$,

$$f(x+n+1) - f(n+1) = f(x) + \log x + \sum_0^{n-1} \log\left(1 + \frac{x}{k+1}\right).$$

Using the convexity of f we obtain

$$\begin{aligned} f(n+2) - f(n+1) &\leq \frac{f(x+n+1) - f(n+1)}{x} \\ &\leq \frac{f(n+3) - f(n+1)}{2} \\ &\leq f(n+3) - f(n+2). \end{aligned}$$

Hence

$$f(x) + \gamma x + \log x + \sum_0^{n-1} \left[\log\left(1 + \frac{x}{k+1}\right) - \frac{x}{k+1} \right]$$

lies between

$$\left[\gamma + \log(n+1) - \sum_0^{n-1} \frac{1}{k+1} \right] x$$

and

$$\left[\gamma + \log(n+2) - \sum_0^{n-1} \frac{1}{k+1} \right] x.$$

We conclude that

$$f(x) = -\gamma x - \log x - \sum_0^{\infty} \left[\log\left(1 + \frac{x}{k+1}\right) - \frac{x}{k+1} \right],$$

that is, $f(x) = \log \Gamma(x)$, $1 < x < 2$, and thereupon that $f(x) = \log \Gamma(x)$, x

positive, thanks to conditions (1) and (2) of the theorem. The theorem follows.

EXERCISES

In the following exercises two famous properties of Γ are to be established with the aid of the theorem of Bohr. They are the *Gauss multiplication formula* and the *Stirling formula*. The idea of using Bohr's theorem for such proofs is due to E. Artin [7, Ch. 3].

1.2. Let n be a given positive integer. Let

$$f(z) = e^{z \log n} \prod_{k=0}^{n-1} \Gamma\left(\frac{z+k}{n}\right).$$

Let $\varphi = \log \circ (f|P)$. Show that $\varphi - \varphi(1)$ fulfills the hypothesis of the theorem of Bohr. Conclude that

$$f = a(n)\Gamma, \quad (1.3)$$

where $a(n) \in P$. The value of $a(n)$ will be determined in Ex. 1.4 below. The formula (1.3) with $a(n)$ so given is the Gauss multiplication formula.

1.3. Show that $\varphi = (x - \frac{1}{2}) \log x - x$ with domain P is convex. Show that if ψ with domain P is convex and satisfies

$$\varphi(x+1) + \psi(x+1) = \log x + \varphi(x) + \psi(x), \quad x \in P,$$

or, equivalently, $\psi(x) = \psi(x+1) + \theta(x)$, where

$$\theta(z) = (z + \frac{1}{2}) \operatorname{Log}\left(1 + \frac{1}{z}\right) - 1,$$

then $\varphi + \psi - \log \circ (\Gamma|P)$ is constant. Show that $\sum_0^\infty \theta(x+n)$ serves for $\psi(x)$. To that end, consider the Laurent expansion of $\theta| \{1 < |z| < +\infty\}$ to establish uniform convergence and with the aid of $\theta''(x) = [2x^2(1+x)^2]^{-1}$ conclude the convexity of $\sum_0^\infty \theta(x+n)$, $x \in P$. We continue with ψ so defined. Determine the Laurent expansion of the second derivative of $\theta| \{1 < |z+1| < +\infty\}$ and conclude, integrating term-by-term, that $\theta(x) < [12x(x+1)]^{-1}$, x positive, and hence that $0 < \psi(x) < (12x)^{-1}$. Show further that $\sum_0^\infty \theta'(x+n)$ converges uniformly in P to $\psi'(x)$ and that $\sum_0^\infty \theta''(x+n)$ converges uniformly in P to $\psi''(x)$.

Let $c \in P$ be such that

$$\Gamma(x) = c \exp[\varphi(x) + \psi(x)], \quad (1.4)$$

x positive. With c determined, we obtain a version of the Stirling formula which states $\lim_{+\infty} \Gamma(x) e^{-\varphi(x)} = c$. Its utility for treating $n!$ for n large is apparent.

1.4. Determination of $a(n)$ and c . Using (1.3) and (1.4) show that

$$a(n) c^{-(n-1)} = \lim_{+\infty} \exp \left[x \log n - \varphi(x) + \sum_{k=0}^{n-1} \varphi\left(\frac{x+k}{n}\right) \right] = \sqrt{n}.$$

Note that $\Gamma(1/2) = \sqrt{\pi}$ (Ex. 3.14, Ch. VIII) and deduce from (1.3), setting $n = 2$ and $z = 1$, that $a(2) = 2\sqrt{\pi}$ and $c = \sqrt{(2\pi)}$. We then have $a(n) = c^{n-1}\sqrt{n}$.

1.5. A Stirling formula in the complex domain. Let $S = \mathbb{C} - \{-t; t \in \mathbb{R}^+\}$. We introduce $\Phi = (z - \frac{1}{2}) \operatorname{Log} z - z$ with domain S and

$$\Psi = \sum_0^{\infty} \theta(z+n) \quad (1.5)$$

with domain S . Both functions are analytic on S and we see from the preceding two exercises that $\Gamma|_S = (2\pi)^{1/2} \exp \circ (\Phi + \Psi)$. The object of the present exercise is to estimate the term Ψ .

Let $0 < \alpha < \pi$. Let $t \in \mathbb{R}^+$ and $|\operatorname{Arg} z| \leq \alpha$. Show that

$$|z+t|^2 \geq (|z|^2 + t^2)[1 + (\cos \alpha)^{-1}].$$

Show that there exists positive A such that

$$|\Psi(z)| \leq A/|z|, \quad |\operatorname{Arg} z| \leq \alpha.$$

Hint: Proceed from (1.5) using the Laurent expansion of $\theta|_{\{1 < |z| < +\infty\}}$ to infer the existence of a positive number B such that $|\theta(z)| \leq B|z|^{-2}$, $|z| > 2$, and noting that for n large and positive we have

$$\{|\operatorname{Arg}(z-n)| \leq \alpha\} \subset \{2 < |z| < +\infty\}.$$

With these facts and the above inequality for $|z+t|^2$ we are led to an estimate of the desired type subject to the restriction $|z| \geq 1$. An estimate of the desired type subject to the condition $|z| < 1$ may be obtained on observing the behavior of $\operatorname{Log}(1+z^{-1})$.

For much more refined results the reader may consult [31] and [130].

We now turn to an integral representation of $\Gamma(z)$, $\operatorname{Re} z > 0$. The starting point is the *Euler integral*

$$\int_0^{+\infty} e^{(z-1) \log t - t} dt, \quad (1.6)$$

which is absolutely convergent when $\operatorname{Re} z > 0$ (that is, the integral obtained by replacing the integrand of (1.6) by its absolute value exists when $\operatorname{Re} z > 0$). We recall that

$$\int_0^{+\infty} t^\alpha e^{-t} dt$$

is convergent for $\alpha > -1$. If $0 < a < b < +\infty$, then $F_{a,b}$ given by

$$F_{a,b}(z) = \int_a^b e^{(z-1) \log t - t} dt \quad (1.7)$$

is entire and differentiation under the integral sign is in effect. Further, on $\{A \leq \operatorname{Re} z \leq B\}$, where $0 < A < B < +\infty$, $F_{a,b}$ tends uniformly to F given by (1.6), that is, given ε positive there exists δ positive such that

$$|F_{a,b}(z) - F(z)| < \varepsilon,$$

$A \leq \operatorname{Re} z \leq B$, $a < \delta$, $b > \delta^{-1}$. Automatically, as a consequence of the fact that $F_{a,b}$ tends to F uniformly in $\{\operatorname{Re} z > 0\}$, we conclude that

$$F'(z) = \int_0^{+\infty} (\log t) e^{(z-1) \log t - t} dt,$$

and

$$F''(z) = \int_0^{+\infty} (\log t)^2 e^{(z-1) \log t - t} dt,$$

$\operatorname{Re} z > 0$. We shall now proceed to show with the aid of the theorem of Bohr that $F(z) = \Gamma(z)$, $\operatorname{Re} z > 0$. It is routine to see that $F(1) = 1$ and that $F(x) > 0$, x positive. To verify that F satisfies the basic functional equation, we apply integration by parts to $F_{a,b}$, obtaining

$$F_{a,b}(z) = e^{z \log b - b} - e^{z \log a - a} - (z-1) F_{a,b}(z) + F_{a,b}(z+1).$$

It is immediate that F satisfies the functional equation of the gamma function.

To show that $\log \circ (F|P)$ is convex, it suffices to show that

$$\left(\frac{F'}{F}\right)'(x) \geq 0,$$

or, equivalently,

$$(FF'' - F'^2)(x) \geq 0, \quad (1.8)$$

x positive. Using the Cauchy-Schwarz-Buniakowsky inequality for integrals (cf. §5, Ch. IV) we have

$$\begin{aligned} [F'_{a,b}(x)]^2 &= \left\{ \int_a^b e^{[(x-1) \log t - t]/2} \cdot \log t \cdot e^{[(x-1) \log t - t]/2} dt \right\}^2 \\ &\leq [F_{a,b}(x)][F''_{a,b}(x)] \end{aligned}$$

and hence we are led to (1.8). By the theorem of Bohr we infer that $F(x) = \Gamma(x)$, x positive, and thereupon by the identity theorem for analytic functions that $F(z) = \Gamma(z)$, $\operatorname{Re} z > 0$.

EXERCISE

1.6. Without reference to the just established fact that $F(z) = \Gamma(z)$, $\operatorname{Re} z > 0$, show on the basis of the functional equation $F(z+1) = z F(z)$ that F is the restriction to $\{\operatorname{Re} z > 0\}$ of a function meromorphic on $\mathbb{C}$. Hint: $F(z+1)/z$ defines a meromorphic function on $\{\operatorname{Re} z > -1\}$ whose restriction to $\{\operatorname{Re} z > 0\}$ is F . The argument is to be repeated inductively. Show that the divisor of the so defined function takes the value -1 at $-n$, $n \in \mathbb{N}$, and elsewhere nonnegative values.

The Hankel formulas. Let $0 < r < +\infty$. We introduce in terms of integrals two functions intimately connected with the gamma function. For $z \in \mathbb{C}$, let

$$\begin{aligned} H_1(z) = & \int_{-\infty}^{-r} \exp[(z-1)(\log |t| - \pi i) + t] dt \\ & + \int_{-\pi}^0 \exp[(z-1)(\log r + i\theta) + re^{i\theta}] ire^{i\theta} d\theta \\ & + \int_0^{\pi} \exp[(z-1)(\log r + i\theta) + re^{i\theta}] ire^{i\theta} d\theta \\ & + \int_{-r}^{-\infty} \exp[(z-1)(\log |t| + \pi i) + t] dt. \end{aligned} \quad (1.9)$$

Clearly,

$$\begin{aligned} H_1(z) = & 2i \sin \pi z \int_r^{+\infty} \exp[(z-1) \log t - t] dt \\ & + \int_{-\pi}^{\pi} \exp[(z-1)(\log r + i\theta) + re^{i\theta}] ire^{i\theta} d\theta. \end{aligned} \quad (1.10)$$

When we interpret the integrals involved in (1.9) in terms of line integrals, the first two in terms of line integrals with integrand $\exp[(z-1)L_1(w) + w]$, where L_1 is the analytic logarithm of the identity map on $\mathbb{C} - \{it: t \in \mathbb{R}^+\}$ satisfying $L_1(1) = 0$, the last two in terms of line integrals with integrand $\exp[(z-1)L_2(w) + w]$, where L_2 is the analytic logarithm of the identity map on $\mathbb{C} - \{-it: t \in \mathbb{R}^+\}$ satisfying $L_2(1) = 0$, we see that $H_1(z)$ is independent of r by virtue of the Cauchy integral theorem.

EXERCISE

1.7. Indicate the paths involved and justify the assertion just made.

Restricting attention to a given z with positive real part, from (1.10) we see on determining $\lim_0 A$, $A(r)$ being the right-hand side of (1.10) for the given value of z , that

$$H_1(z) = 2i \sin \pi z \Gamma(z). \quad (1.11)$$

Since both sides define functions analytic on $\mathbb{C}$, we see that (1.11) holds throughout $\mathbb{C}$, due regard being given to the poles of Γ .

EXERCISE

1.8. We define H_2 similarly by

$$\begin{aligned}
 2\pi i H_2(z) = & \int_{-\infty}^{-r} \exp[(-z)(\log |t| - \pi i) + t] dt \\
 & + \int_{-\pi}^0 \exp[(-z)(\log r + i\theta) + re^{i\theta}]ire^{i\theta} d\theta \\
 & + \int_0^{\pi} \exp[(-z)(\log r + i\theta) + re^{i\theta}]ire^{i\theta} d\theta \\
 & + \int_{-r}^{-\infty} \exp[(-z)(\log |t| + \pi i) + t] dt.
 \end{aligned} \tag{1.12}$$

Show that

$$\Gamma H_2 = 1_{\mathbb{C}}. \tag{1.13}$$

Formulas (1.11) and (1.13) are known as the *Hankel formulas*. Hint: Parallel the argument leading to (1.11), bringing into play Ex. 3.14, Ch. VIII.

2. THE RIEMANN ZETA FUNCTION AND ITS FUNCTIONAL EQUATION

The Riemann zeta function appeared in Ex. 3.10, Ch. VIII. In the first instance, we are concerned with $\sum_1^{\infty} e^{-z \log n}$ with domain $\{\operatorname{Re} z > 1\}$. We speak provisionally because we shall obtain relations permitting us to infer that the function so specified is the restriction to $\{\operatorname{Re} z > 1\}$ of a function meromorphic on $\mathbb{C}$. The latter function will be termed the *Riemann zeta function* and will be denoted by ζ .

Let us first consider the function

$$F = \int_0^{+\infty} \frac{e^{z \log t}}{e^t - 1} dt$$

with domain $\{\operatorname{Re} z > 0\}$. The integral is absolutely convergent for such z . Using the identity

$$\frac{1}{e^t - 1} = \sum_1^n e^{-kt} + \frac{e^{-(n+1)t}}{1 - e^{-t}},$$

we obtain

$$F(z) = \sum_1^n \int_0^{+\infty} e^{(z \log t - kt)} dt + \int_0^{+\infty} \frac{e^{[z \log t - (n+1)t]}}{1 - e^{-t}} dt. \tag{2.1}$$

Making the change of variable carrying t into t/k , we find

$$\int_0^{+\infty} e^{(z \log t - kt)} dt = \Gamma(z+1) e^{-(z+1) \log k}. \quad (2.2)$$

Given ε positive, with $x = \operatorname{Re} z$, we see that the absolute value of the last integral on the right-hand side of (2.1) is less than

$$\int_0^\varepsilon \frac{e^{x \log t}}{1 - e^{-t}} dt + (1 - e^{-\varepsilon})^{-1} \frac{\Gamma(x+1)}{(n+1)^{x+1}}.$$

The estimates involved are straightforward, the second factor of the second term arising from an integral of the form occurring in (2.2), k being replaced by $(n+1)$. We conclude that

$$\limsup_{n \rightarrow \infty} \left| F(z) - \Gamma(z+1) \sum_1^n e^{-(z+1) \log k} \right| \leq \int_0^\varepsilon \frac{e^{x \log t}}{1 - e^{-t}} dt,$$

and hence by the arbitrariness of ε that

$$F(z) = \Gamma(z+1) \sum_1^\infty e^{-(z+1) \log k}. \quad (2.3)$$

We now introduce a function defined by an integral, as we did the functions H_1 and H_2 , but this time a parameter that will be significant enters. Given positive ρ , *not* an integer multiple of 2π , we introduce

$$\begin{aligned} H^\rho(z) = & \int_{-\infty}^{-\rho} \frac{e^{z(\log |t| - \pi i)}}{1 - e^{-t}} dt + \int_{-\pi}^\pi \frac{e^{z(\log \rho + i\theta)}}{1 - \exp(-\rho e^{i\theta})} i\rho e^{i\theta} d\theta \\ & + \int_{-\rho}^{-\infty} \frac{e^{z(\log |t| + \pi i)}}{1 - e^{-t}} dt. \end{aligned}$$

On combining the first and third integrals, we obtain

$$H^\rho(z) = 2i \sin \pi z \int_\rho^{+\infty} \frac{e^{z \log t}}{e^t - 1} dt + \int_{-\pi}^\pi \frac{e^{z(\log \rho + i\theta)}}{1 - \exp(-\rho e^{i\theta})} i\rho e^{i\theta} d\theta. \quad (2.4)$$

Now the first integral in (2.4) is readily seen to define an entire function. We introduce

$$G(z) = \int_1^{+\infty} \frac{e^{z \log t}}{e^t - 1} dt.$$

Clearly

$$|G(z)| = \left| \int_1^{+\infty} \frac{e^{z \log t - t}}{1 - e^{-t}} dt \right| \leq (1 - e^{-1})^{-1} \Gamma(|z| + 1). \quad (2.5)$$

We proceed to obtain an upper estimate for the order of the entire function G and thereupon of H^ρ . We can use the Stirling formula to control $M(r; G)$. Actually a much cruder estimate of $\Gamma|P$ is adequate. From the convexity of $\log \circ (\Gamma|P)$, the nonconstancy of Γ , and the fact that $\Gamma(1) = \Gamma(2)$, we know that Γ is increasing on $[2, +\infty[$. We obtain the crude inequality

$$\Gamma(x) < \Gamma([x] + 1) < [x]^{[x]} \leq e^{x \log x},$$

$[x] = \max\{k: k \in \mathbf{Z}, k \leq x\}$, for $x \geq 2$. Using this estimate and (2.5) we obtain

$$\log M(r; G) \leq -\log(1 - e^{-1}) + (r + 1) \log(r + 1),$$

$r \geq 1$, and hence $\alpha(G) \leq 1$. It is easy to see that $\alpha(H^\rho - 2i \sin \pi z G) \leq 1$. Hence we conclude $\alpha(H^\rho) \leq 1$.

Interpreting the integrals involved in the definition of H^ρ as line integrals, as we did in the case of H_1 , we see with the aid of the Cauchy integral theorem that $H^\rho(z)$ is independent of ρ , $0 < \rho < 2\pi$. Let $H(z)$ denote the common value. We fix z , $\operatorname{Re} z > 0$, and obtain, using (2.3), (2.4), and the obvious $H(z) = \lim_0 H^\rho(z)$, the important relation

$$H(z) = 2i \sin \pi z \Gamma(z + 1) \sum_1^\infty e^{-(z+1) \log k}. \quad (2.6)$$

From (2.6) we reach the basic conclusion that $\sum_1^\infty e^{-z \log k}$ with domain $\{\operatorname{Re} z > 1\}$ is the restriction of a function meromorphic on $\mathbf{C}$, which function, ζ , satisfies the relation

$$\zeta(z) = -\frac{H(z-1)}{2i \sin \pi z \Gamma(z)}, \quad (2.7)$$

the usual conventions holding.

EXERCISE

2.1. Show with the aid of (2.7) that $\partial_\zeta(1) = -1$, $\partial_\zeta(z) \geq 0$, $z \neq 1$. Show from (2.4) that $H(0) = 2\pi i$ and conclude that $\operatorname{res}(1; \zeta) = 1$.

We consider now $H^{(2n+1)\pi}$, $n \in \mathbf{N}$. From (2.4) we see that for fixed z , $\operatorname{Re} z < -1$, we have $\lim_{n \rightarrow \infty} H^{(2n+1)\pi}(z) = 0$. It is to be noted that

$$0 < \inf \left\{ |1 - e^z| : z \in \bigcup_{n \in \mathbf{N}} C[0; (2n+1)\pi] \right\}.$$

Further, by virtue of the line integral interpretations and the residue theorem,

$H^{(2n+1)\pi}(z) - H(z)$ is simply $2\pi i$ times the sum of the residues of the meromorphic function

$$\frac{e^{z \log w}}{1 - e^{-w}}$$

with domain $\mathbf{C} - \{-t: t \in \mathbf{R}^+\}$ at the points $2\pi i k$, $1 \leq |k| \leq n$, $k \in \mathbf{Z}$. That is,

$$H^{(2n+1)\pi}(z) - H(z) = 4\pi i \cos \frac{\pi z}{2} \sum_{k=1}^n e^{z \log(2\pi k)}.$$

Hence we see that for $\operatorname{Re} z < -1$ we have

$$H(z) = -4\pi i \cos \frac{\pi z}{2} e^{z \log 2\pi} \zeta(-z),$$

and hence for $\operatorname{Re} z < 0$ we have

$$H(z-1) = -2i \sin \frac{\pi z}{2} e^{z \log 2\pi} \zeta(1-z).$$

By appeal to (2.7) and the identity theorem for meromorphic functions we obtain the *functional equation for the Riemann zeta function*

$$\zeta(1-z) = 2 \cos \frac{\pi z}{2} e^{-z \log 2\pi} \Gamma(z) \zeta(z), \quad (2.8)$$

valid throughout $\mathbf{C}$ with the usual conventions prevailing.

This important relation yields information on the zeros of the Riemann zeta function. Thanks to the Euler product formula for $\zeta(z)$, $\operatorname{Re} z > 1$, of Ex. 3.10, Ch. VIII, we see that ζ has no zeros in $\{\operatorname{Re} z > 1\}$. Using (2.8), the fact that Γ has no zeros, and our knowledge of the zeros of the cosine, we see that $\partial_\zeta(z) = 1$, $z \in T = \{-(2+2k): k \in \mathbf{N}\}$ and $\partial_\zeta(z) = 0$, $\operatorname{Re} z < 0$, $z \notin T$. The members of T are termed the “*trivial*” zeros of ζ . We conclude: *the nontrivial zeros of ζ lie in the strip $\{0 \leq \operatorname{Re} z \leq 1\}$.*†

We introduce two traditional auxiliary entire functions. The first, ξ , is to satisfy

$$\xi(z) = \frac{1}{2} z(z-1) e^{-(z \log \pi)/2} \Gamma(z/2) \zeta(z), \quad z \neq -2k, \quad k \in \mathbf{N}. \quad (2.9)$$

From (2.7), the fact that $\alpha(H) \leq 1$, and Ex. 6.4, Ch. VIII, we see that $\alpha(\xi) \leq 1$. We shall now show that $\alpha(\xi)$ is *exactly* 1. Indeed for $r > 2$,

$$\xi(r) \geq e^{-(r \log \pi)/2} \Gamma(r/2).$$

† Note that the term “strip” is used with a certain amount of license.

Suppose that $y > 2$. Using the convexity of $\log \circ (\Gamma | \mathbf{P})$ we have

$$\frac{\log \Gamma(x) - \log \Gamma(2)}{x - 2} \geq \frac{\log \Gamma(y) - \log \Gamma(2)}{y - 2}$$

for $x \geq y$. Hence for $r \geq 2y$ we have

$$\log \xi(r) \geq -\frac{r}{2} \log \pi + \frac{\log \Gamma(y)}{y - 2} \left(\frac{r}{2} - 2 \right).$$

It follows from the fact that $\lim_{+\infty} [\log \Gamma(y)]/y = +\infty$ that $\alpha(\xi) = 1$.[†] By a straightforward calculation using the functional equation of the Riemann zeta function, the Gauss multiplication formula (Exs. 1.2 and 1.4, this chapter), and Ex. 3.14(b), Ch. VIII, we see that

$$\xi(1 - z) = \xi(z). \quad (2.10)$$

EXERCISE

2.2. Verify the relation (2.10). Hint: Introduce $A(z) = \zeta(1 - z)/\zeta(z)$ and $B(z) = \xi(z)/\zeta(z)$. The identity (2.10) may be reduced to $A(z) = B(z)/B(1 - z)$.

We now introduce the second traditional auxiliary function Ξ given by $\Xi(z) = \xi(\frac{1}{2} + iz)$ and we see that Ξ is an even function of order 1. Consequently, Ξ has infinitely many zeros and hence so does ξ . Now the zeros of ξ are precisely the nontrivial zeros of ζ as we see from (2.9). Hence we obtain the following well-known property of the Riemann zeta function.

Theorem 2.1: ζ has infinitely many zeros in the strip $\{0 \leq \operatorname{Re} z \leq 1\}$.

This theorem brings us to the most celebrated conjecture of modern analysis—the *hypothesis of Riemann* that all the nontrivial zeros of ζ lie on $\{\operatorname{Re} z = \frac{1}{2}\}$. It is not possible to give a notion of the wealth of work dedicated to the Riemann zeta function and its zeros in a few lines. The reader is referred to the treatise of Titchmarsh [118] for an authoritative account of the subject. We do mention, however, two important classical results:

- (1) $\zeta(z) \neq 0$, $\operatorname{Re} z = 1$ (Hadamard and de la Vallée Poussin).
- (2) ζ has infinitely many zeros on $\{\operatorname{Re} z = \frac{1}{2}\}$ (Hardy).

The first of these will be proved in the next section.

[†] We even have the more precise relation $\lim_{+\infty} [\log^+ \log^+ M(r; \xi)]/\log r = 1$.

3. THE THEOREM OF HADAMARD AND DE LA VALLÉE POUSSIN ON THE ZEROS OF THE RIEMANN ZETA FUNCTION

Starting with the product representation of $\zeta(z)$, $\operatorname{Re} z > 1$, of Ex. 3.10, Ch. VIII, we see on taking logarithmic derivatives that

$$-\frac{\zeta'}{\zeta}(z) = \sum_0^{\infty} \frac{(\log p_k) e^{-z \log p_k}}{1 - e^{-z \log p_k}}, \quad \operatorname{Re} z > 1.$$

We introduce Λ with domain the set of positive integers satisfying the condition $\Lambda(p^m) = \log p$ where p is a positive prime and m is a positive integer and $\Lambda(n) = 0$ when n is not of this form. It is readily verified that if A is the cartesian product of the set of positive primes and the set of positive integers, and $\alpha(p, n) = 1$ when $\log n / \log p$ is a positive integer and otherwise $\alpha(p, n) = 0$, $(p, n) \in A$, then

$$(\log p) \alpha(p, n) e^{-z \log n} \quad (3.1)$$

is summable on A in the sense of §4, Ch. VIII, for each z satisfying $\operatorname{Re} z > 1$, since for each positive prime p we have

$$\frac{e^{-z \log p}}{1 - e^{-z \log p}} = \sum_2^{\infty} \alpha(p, n) e^{-z \log n}, \quad \operatorname{Re} z > 0.$$

We conclude that

$$-\frac{\zeta'}{\zeta}(z) = \sum_2^{\infty} \Lambda(n) e^{-z \log n}, \quad \operatorname{Re} z > 1. \quad (3.2)$$

EXERCISE

3.1. Establish the summability of (3.1) on A and the relation (3.2) on the basis of the developments of §4, Ch. VIII.

It will be convenient to treat first the following lemma: Given $f: X \times Y \rightarrow \mathbb{C}$ summable. For each $y \in Y$, let $g_y(x) = f(x, y)$, g_y having domain X . Then each g_y is summable, $S(g_y)$, $y \in Y$, is summable, and

$$\sum_{y \in Y} S(g_y) = \sum_{(x, y) \in X \times Y} f(x, y).$$

We now establish the theorem of Hadamard and de la Vallée Poussin concerning the zeros of the Riemann zeta function.

Theorem 3.1: ζ has no zeros on $\{\operatorname{Re} z = 1\}$.

PROOF: The extremely ingenious proof starts by introducing the function

$$\Phi = \zeta^3(\zeta \circ \tau_{it})^4(\zeta \circ \tau_{2it})$$

where $\tau_a = z + a$, $z \in \mathbf{C}$, and $t \in \mathbf{R} - \{0\}$. Suppose that $\zeta(1 + it) = 0$. Then Φ has a zero at 1 since ζ has a pole of the first order at 1. Hence $\operatorname{Re}[\Phi'(x)/\Phi(x)] > 0$, $1 < x < a$, for some a satisfying $a > 1$. By (3.2)

$$\frac{\Phi'}{\Phi}(z) = - \sum_2^{\infty} \Lambda(n) e^{-z \log n} (3 + 4e^{-it \log n} + e^{-2it \log n})$$

and hence

$$\operatorname{Re} \frac{\Phi'}{\Phi}(x) = -2 \sum_2^{\infty} \Lambda(n) e^{-x \log n} [\cos(t \log n) + 1]^2 \leq 0$$

when $x > 1$. The contradiction is manifest. Consequently ζ has no zero on $\{\operatorname{Re} z = 1\}$ and the theorem is established.

4. THE PRIME NUMBER THEOREM OF HADAMARD AND DE LA VALLÉE POUSSIN

The exposition of this section follows substantially N. Wiener's proof of the Prime Number Theorem in the form given by Landau [141]. That there is an intimate connection between the Riemann zeta function and the set of positive primes is, of course, clear from the Euler product formula of Ex. 3.10, Ch. VIII. The point is to take advantage of the relation. An important link is furnished by the function $\psi(x) = \sum_{n \leq x} \Lambda(n)$ with domain $\mathbf{P}$, Λ retaining the meaning given to it in the preceding section of this chapter. Let $3 \leq x < +\infty$. Let p be a positive prime. Let

$$m(p) = \max\{k: k \in \mathbf{N}, p^k \leq x\}.$$

Clearly $m(p) \leq (\log x)/\log p$. Hence

$$\psi(x) = \sum m(p) \log p \leq \pi(x) \log x, \quad (4.1)$$

the sum being taken over the set of positive primes. When $1 < y < x$,

$$\pi(x) \leq \pi(y) + \sum \frac{\log p}{\log y} \leq \pi(y) + \frac{\psi(x)}{\log y},$$

the sum being taken over the set of primes in $]y, x]$. Now setting

$y = x(\log x)^{-2}$, which satisfies the above condition, and noting that $\pi(y) < y$, we obtain

$$\pi(x) \frac{\log x}{x} < \frac{1}{\log x} + \frac{\psi(x)}{x} \frac{\log x}{\log x - 2 \log(\log x)}. \quad (4.2)$$

By (4.1) and (4.2) we infer that *the Prime Number Theorem is equivalent to the assertion*

$$\lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = 1. \quad (4.3)$$

A link has been established between the Prime Number Theorem and a property of the Riemann zeta function.

In order to execute the proof of (4.3) we shall want to have two ancillary theorems available.

The first is the following form of the Riemann–Lebesgue lemma for C' functions. It states:

Let $-\infty < a < b < +\infty$. Let $f: [a, b] \rightarrow \mathbf{C}$ possess a derivative at each point of $[a, b]$ (one sided at a and at b), f' being continuous at each point of $[a, b]$.† Let $M_0 = \max_{[a, b]} |f|$, $M_1 = \max_{[a, b]} |f'|$. Let

$$g(\alpha) = \int_a^b f(x) e^{i\alpha x} dx.$$

Then

$$|\alpha g(\alpha)| \leq 2M_0 + M_1(b - a), \quad \alpha \in \mathbf{R}.$$

The proof is immediate on integrating by parts. We find

$$i\alpha g(\alpha) = f(b) e^{i\alpha b} - f(a) e^{i\alpha a} - \int_a^b f'(x) e^{i\alpha x} dx$$

and hence

$$|\alpha g(\alpha)| \leq |f(a)| + |f(b)| + \int_a^b |f'(x)| dx.$$

The asserted inequality is now evident.

The second result that we shall want is a very special instance of a Fubini theorem. It states:

Let $-\infty < a < b < +\infty$, $-\infty < c < d < +\infty$. Let $f: [a, b] \rightarrow \mathbf{C}$ satisfy $V[f] < +\infty$. Let g be a continuous map of $[a, b] \times [c, d]$ into $\mathbf{C}$. Then

† That is, let $f \in C'[a, b]$.

$$\int_c^d \left[\int_a^b f(x) g(x, y) dx \right] dy = \int_a^b \left[\int_c^d f(x) g(x, y) dy \right] dx, \quad (4.4)$$

the integrals involved existing.†

The proof will follow from Ex. 4.1 below.

EXERCISE

4.1. (a) With f as in the above-stated result and h a continuous map of $[a, b]$ into $\mathbb{C}$, fh possesses an integral in the sense of Riemann and

$$\int_a^b fh dx = \int_a^b f dH,$$

where

$$H(x) = \int_a^x h(t) dt.$$

(b) The function given by

$$\int_a^b f(x) g(x, y) dx$$

is continuous on $[c, d]$. (c) Let n be a positive integer and approximate the integral on the left side of (4.4) by a sum of the form

$$\sum_0^{n-1} \left[\int_a^b f(x) g(x, y_k) dx \right] (y_{k+1} - y_k) = \int_a^b \left[\sum_0^{n-1} f(x) g(x, y_k) (y_{k+1} - y_k) \right] dx,$$

where $y_k = c + n^{-1}k(d - c)$, $k = 0, \dots, n$. Note that

$$\left| \int_c^d f(x) g(x, y) dy - \sum_0^{n-1} f(x) g(x, y_k) (y_{k+1} - y_k) \right| \leq M \omega_g[n^{-1}(d - c)](d - c),$$

where $M = \sup_{[a, b]} |f|$ and conclude (4.4).

We now turn to the proof of (4.3) to which, as we have seen, the Prime Number Theorem reduces. We make a change of variable for convenience introducing $K(u) = \psi(e^u) e^{-u}$, $u \in \mathbb{R}$, and note that (4.3) is equivalent to

$$\lim_{+\infty} K = 1. \quad (4.5)$$

† We are not presupposing a knowledge of such results, though to be sure some readers are familiar with fairly general forms of the Fubini theorem. It is worth emphasizing the important role played in analysis by theorems that permit interchange of order of integration or summation.

We introduce the function G meromorphic on $\mathbb{C}$, satisfying

$$G(z) = - \left[\frac{\zeta'}{\zeta}(z) + \frac{z}{z-1} \right] \frac{1}{z}$$

when $z \neq 0, 1$. For $\operatorname{Re} z > 1$ we have by (3.2) of the preceding section

$$G(z) = \frac{1}{z} \int_1^{+\infty} e^{-z \log t} d\psi(t) - \frac{1}{z-1},$$

and hence, by the change of variable $t = e^u$,

$$G(z) = \frac{1}{z} \int_0^{+\infty} e^{-zu} d\psi(e^u) - \frac{1}{z-1}.$$

Integrating by parts and noting that $\psi(x) \leq x \log x$, $x \geq 1$, we obtain

$$\begin{aligned} G(z) &= \int_0^{+\infty} \psi(e^u) e^{-zu} du - \frac{1}{z-1} \\ &= \int_0^{+\infty} [K(u) - 1] e^{-(z-1)u} du. \end{aligned}$$

We note that G is analytic at each point of $\{\operatorname{Re} z \geq 1\}$ since ζ has no zeros on $\{\operatorname{Re} z \geq 1\}$ (cf. Theorem 3.1, this chapter) and has a pole only at 1, which is harmless as far as G is concerned. By the above form of the Riemann-Lebesgue lemma applied on $[-2, 0]$ and $[0, 2]$ we see that for each positive λ there exists a positive number $C(\lambda)$ such that

$$\left| \int_{-2}^2 G(1 + \varepsilon + i\lambda t) \frac{\lambda}{2} \left(1 - \frac{|t|}{2}\right) e^{i(\lambda y)t} dt \right| \leq \frac{C(\lambda)}{y} \quad (4.6)$$

for y positive, $0 \leq \varepsilon \leq 1$.

Let

$$G_n(z) = \int_0^{n+1} [K(u) - 1] e^{-(z-1)u} du, \quad n \in \mathbb{N}.$$

Applying the ancillary theorem of Fubini type we obtain

$$\begin{aligned} &\int_{-2}^2 G_n(1 + \varepsilon + i\lambda t) \frac{\lambda}{2} \left(1 - \frac{|t|}{2}\right) e^{i(\lambda y)t} dt \\ &= \int_0^{n+1} \left\{ [K(u) - 1] e^{-\varepsilon u} \int_{-2}^2 \frac{\lambda}{2} \left(1 - \frac{|t|}{2}\right) e^{i\lambda(y-u)t} dt \right\} du. \end{aligned} \quad (4.7)$$

It is to be observed that no restriction need be imposed on ε, λ, y in (4.7). We

continue assuming λ, y positive and $0 < \varepsilon \leq 1$. It is routine to see that

$$\int_{-2}^2 \frac{\lambda}{2} \left(1 - \frac{|t|}{2}\right) e^{i\lambda(y-u)t} dt = \lambda \left[\frac{\sin \lambda(u-y)}{\lambda(u-y)} \right]^2,$$

when $u \neq y$ (and is $=\lambda$ when $u = y$). Introducing the change of variable $u = y + \lambda^{-1}v$, we see that each side of (4.7) is equal to

$$\int_{-\lambda y}^{\lambda(n+1-y)} [K(y + \lambda^{-1}v) - 1] e^{-\varepsilon(y+\lambda^{-1}v)} \left(\frac{\sin v}{v}\right)^2 dv. \quad (4.8)$$

We note that G_n tends uniformly to G in $\{\operatorname{Re} z > 1\}$. Hence by (4.6), (4.7), and (4.8) we are led to

$$\left| \int_{-\lambda y}^{+\infty} [K(y + \lambda^{-1}v) - 1] e^{-\varepsilon(y+\lambda^{-1}v)} \left(\frac{\sin v}{v}\right)^2 dv \right| \leq \frac{C(\lambda)}{y}$$

and, since $K(u) = 0$ for u negative,

$$\left| \int_{-\infty}^{+\infty} K(y + \lambda^{-1}v) e^{-\varepsilon(y+\lambda^{-1}v)} \left(\frac{\sin v}{v}\right)^2 dv - \int_{-\lambda y}^{+\infty} e^{-\varepsilon(y+\lambda^{-1}v)} \left(\frac{\sin v}{v}\right)^2 dv \right| \leq \frac{C(\lambda)}{y}. \quad (4.9)$$

We next show that (4.9) remains valid when ε is replaced by 0. Indeed, when $0 < \varepsilon \leq 1$, from (4.9) we obtain

$$\int_{-\infty}^s K(y + \lambda^{-1}v) e^{-\varepsilon(y+\lambda^{-1}v)} \left(\frac{\sin v}{v}\right)^2 dv \leq \int_{-\infty}^{+\infty} \left(\frac{\sin v}{v}\right)^2 dv + \frac{C(\lambda)}{y}, \quad (4.10)$$

$-\infty < s \leq +\infty$. Clearly (4.10) remains valid with 0 replacing ε . We have

$$\begin{aligned} \int_{-\infty}^s K(y + \lambda^{-1}v) \left(\frac{\sin v}{v}\right)^2 dv &\leq \lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{+\infty} e^{-\varepsilon(y+\lambda^{-1}v)} K(y + \lambda^{-1}v) \left(\frac{\sin v}{v}\right)^2 dv \\ &\leq \int_{-\infty}^{+\infty} K(y + \lambda^{-1}v) \left(\frac{\sin v}{v}\right)^2 dv, \end{aligned}$$

y and λ fixed, $s \in \mathbf{R}$. Hence

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{+\infty} e^{-\varepsilon(y+\lambda^{-1}v)} K(y + \lambda^{-1}v) \left(\frac{\sin v}{v}\right)^2 dv = \int_{-\infty}^{+\infty} K(y + \lambda^{-1}v) \left(\frac{\sin v}{v}\right)^2 dv.$$

Similarly we have

$$\lim_{\varepsilon \rightarrow 0} \int_{-\lambda y}^{+\infty} e^{-\varepsilon(y+\lambda^{-1}v)} \left(\frac{\sin v}{v}\right)^2 dv = \int_{-\lambda y}^{+\infty} \left(\frac{\sin v}{v}\right)^2 dv,$$

and we conclude that (4.9) remains valid when ε is replaced by 0.

Let $\varphi: \mathbf{P} \rightarrow \mathbf{P}$ and satisfy $\lim_{+\infty} \varphi = +\infty$, $\lim_{+\infty} \varphi(\lambda)/\lambda = 0$, for example, let $\varphi(\lambda) = \sqrt{\lambda}$ —the exact choice is unimportant. Since $0 \leq \psi(a) \leq \psi(b)$, $0 < a < b < +\infty$, we have

$$K[y - \lambda^{-1} \varphi(\lambda)]e^{y - \lambda^{-1} \varphi(\lambda)} \leq K(y + \lambda^{-1} v)e^{y + \lambda^{-1} \varphi(\lambda)} \quad (4.11)$$

and

$$K[y + \lambda^{-1} \varphi(\lambda)]e^{y + \lambda^{-1} \varphi(\lambda)} \geq K(y + \lambda^{-1} v)e^{y - \lambda^{-1} \varphi(\lambda)} \quad (4.12)$$

when $-\varphi(\lambda) \leq v \leq \varphi(\lambda)$. By (4.11) and (4.9) with 0 replacing ε , we obtain

$$\begin{aligned} K[y - \lambda^{-1} \varphi(\lambda)] & \int_{-\varphi(\lambda)}^{\varphi(\lambda)} \left(\frac{\sin v}{v} \right)^2 dv \\ & \leq e^{2\lambda^{-1} \varphi(\lambda)} \int_{-\varphi(\lambda)}^{\varphi(\lambda)} K(y + \lambda^{-1} v) \left(\frac{\sin v}{v} \right)^2 dv \\ & \leq e^{2\lambda^{-1} \varphi(\lambda)} \left[\int_{-\lambda y}^{+\infty} \left(\frac{\sin v}{v} \right)^2 dv + \frac{C(\lambda)}{y} \right]. \end{aligned} \quad (4.13)$$

By (4.13) for each positive λ we have

$$\limsup_{+\infty} K \leq e^{2\lambda^{-1} \varphi(\lambda)} \int_{-\infty}^{+\infty} \left(\frac{\sin v}{v} \right)^2 dv / \int_{-\varphi(\lambda)}^{\varphi(\lambda)} \left(\frac{\sin v}{v} \right)^2 dv, \quad (4.14)$$

and hence, taking the limit of the right-hand side at $+\infty$, we have

$$\limsup_{+\infty} K \leq 1.$$

We see, in particular, that there exists M positive such that $K(u) \leq M$, $u \in \mathbf{R}$.

We now consider $\liminf_{+\infty} K$.† By (4.12) and (4.9) with 0 replacing ε we have

$$\begin{aligned} K[y + \lambda^{-1} \varphi(\lambda)] & \int_{-\varphi(\lambda)}^{\varphi(\lambda)} \left(\frac{\sin v}{v} \right)^2 dv \\ & \geq e^{-2\lambda^{-1} \varphi(\lambda)} \int_{-\varphi(\lambda)}^{\varphi(\lambda)} K(y + \lambda^{-1} v) \left(\frac{\sin v}{v} \right)^2 dv \\ & = e^{-2\lambda^{-1} \varphi(\lambda)} \left[\int_{-\infty}^{+\infty} K(y + \lambda^{-1} v) \left(\frac{\sin v}{v} \right)^2 dv - \int_{-\infty}^{-\varphi(\lambda)} - \int_{\varphi(\lambda)}^{+\infty} \right] \\ & \geq e^{-2\lambda^{-1} \varphi(\lambda)} \left[\int_{-\lambda y}^{+\infty} \left(\frac{\sin v}{v} \right)^2 dv - \frac{C(\lambda)}{y} - \frac{2M}{\varphi(\lambda)} \right]. \end{aligned} \quad (4.15)$$

† The definition is dual to that of $\limsup$, “ $\inf$ ” and “ $\sup$ ” replacing one another.

It is to be understood that on the third line the integrand is the same for each integral. The term $2M/\varphi(\lambda)$ on the last line arises from the crude estimate obtained when the bound M is used and $(\sin v)^2$ is replaced by 1. By (4.15) for each positive λ we have

$$\liminf_{+\infty} K \geq e^{-2\lambda - 1/\varphi(\lambda)} \left[\int_{-\infty}^{+\infty} \left(\frac{\sin v}{v} \right)^2 dv - \frac{2M}{\varphi(\lambda)} \right] / \int_{-\varphi(\lambda)}^{\varphi(\lambda)} \left(\frac{\sin v}{v} \right)^2 dv, \quad (4.16)$$

and hence, taking the limit of the right-hand side at $+\infty$, we have

$$\liminf_{+\infty} K \geq 1.$$

It follows from the definitions of limit superior and limit inferior that

$$\lim_{+\infty} K = 1.$$

The proof of the Prime Number Theorem is complete.

An essential point of the proof is the *order* of the limit processes which freed us from the term $C(\lambda)$ in (4.14) and (4.16). We also remark that no use was made in the proof of the exact value of

$$\int_{-\infty}^{+\infty} \left(\frac{\sin v}{v} \right)^2 dv$$

(which is π as one may show with the aid of the Cauchy theorem).